

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session e7e8ae95 · Data hash 83e3...7a3b

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 15:58 and 16:55 UTC. Reported results: first_step_loss 1.89042, final_loss_mean_last100 7.22038. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	e7e8ae95-7078-4cd7-95da-d1c32ed31ed3
Initialized	2026-10-07T15:58:52Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	8243b354d3567105912cea2123a6200c19b0b46e8341e345c1f59213b5cd6303
Disclosure	open
Source code, run 1	7967595a237fd59555109b9e436078b63e9bee1a88069893cb5d818c5fcabcbe

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	56 minutes 32 secon...	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
train_loss	7.4928	3235 (131 readings)	explicit
steps	3235		explicit
peak_memory_reserved_gb	5.32812		explicit
mean_tokens_per_second	3028.81		explicit

Inline claim	Value	Phase	Output line
first_step_loss	1.89042		138
final_loss_mean_last100	7.22038		139

Seed committed: seed:42, output line 1, 2026-10-07T15:58:53Z.

4 Execution evidence

Coherence

Coherent (0.38): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	3395.2 s	3.01 GB	80%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 727 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	15:58:53	7467a68a635fb44e	727
2	run_end	16:55:25	7467a68a635fb44e	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 728 files read, 0 files written, 2 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T16:55:27.102054+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	83e379ad2c69e034a106600a4146911d1837488067750279186839bc636e7a3b
Code bundle hash	cf2c2c93a72bd28c8f756591cfc27cae1df2addacd52a2e4d13c8669c4680ec4

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
train_loss	1.89042	1	4	explicit
train_loss	1.09076	25	5	explicit
train_loss	1.18483	50	6	explicit
train_loss	1.24726	75	7	explicit
train_loss	9.95975	100	8	explicit
train_loss	9.70156	125	9	explicit
train_loss	9.36604	150	10	explicit
train_loss	8.57735	175	11	explicit
train_loss	7.8647	200	12	explicit
train_loss	7.55165	225	13	explicit
train_loss	7.50826	250	14	explicit
train_loss	7.44696	275	15	explicit
train_loss	7.58703	300	16	explicit
train_loss	7.21846	325	17	explicit
train_loss	7.39516	350	18	explicit
train_loss	7.05618	375	19	explicit
train_loss	7.34683	400	20	explicit
train_loss	7.25541	425	21	explicit
train_loss	7.38498	450	22	explicit
train_loss	7.11201	475	23	explicit
train_loss	7.21844	500	24	explicit
train_loss	7.30216	525	25	explicit
train_loss	7.35955	550	26	explicit
train_loss	7.41517	575	27	explicit
train_loss	7.39006	600	28	explicit
train_loss	7.393	625	29	explicit
train_loss	7.29203	650	30	explicit
train_loss	7.37698	675	31	explicit
train_loss	7.25098	700	32	explicit
train_loss	7.073	725	33	explicit
train_loss	7.19104	750	34	explicit
train_loss	7.12901	775	35	explicit
train_loss	7.55612	800	36	explicit
train_loss	7.43066	825	37	explicit
train_loss	7.15925	850	38	explicit
train_loss	7.16717	875	39	explicit
train_loss	7.14667	900	40	explicit
train_loss	7.3102	925	41	explicit
train_loss	7.14592	950	42	explicit
train_loss	7.18849	975	43	explicit
train_loss	7.18832	1000	44	explicit
train_loss	7.51103	1025	45	explicit
train_loss	7.06494	1050	46	explicit
train_loss	7.11098	1075	47	explicit

Metric	Value	Step	Line	Source
train_loss	7.1793	1100	48	explicit
train_loss	7.61247	1125	49	explicit
train_loss	7.44132	1150	50	explicit
train_loss	7.33797	1175	51	explicit
train_loss	7.16563	1200	52	explicit
train_loss	7.24373	1225	53	explicit
train_loss	7.30286	1250	54	explicit
train_loss	7.269	1275	55	explicit
train_loss	7.31434	1300	56	explicit
train_loss	7.14747	1325	57	explicit
train_loss	7.22077	1350	58	explicit
train_loss	7.25226	1375	59	explicit
train_loss	7.19525	1400	60	explicit
train_loss	7.10378	1425	61	explicit
train_loss	7.43932	1450	62	explicit
train_loss	7.24119	1475	63	explicit
train_loss	7.49699	1500	64	explicit
train_loss	7.31041	1525	65	explicit
train_loss	7.20141	1550	66	explicit
train_loss	7.3733	1575	67	explicit
train_loss	7.09624	1600	68	explicit
train_loss	7.41737	1625	69	explicit
train_loss	7.17952	1650	70	explicit
train_loss	6.99467	1675	71	explicit
train_loss	7.18619	1700	72	explicit
train_loss	7.10642	1725	73	explicit
train_loss	7.3817	1750	74	explicit
train_loss	7.2934	1775	75	explicit
train_loss	7.21804	1800	76	explicit
train_loss	7.44726	1825	77	explicit
train_loss	7.33521	1850	78	explicit
train_loss	7.24261	1875	79	explicit
train_loss	7.09905	1900	80	explicit
train_loss	7.52602	1925	81	explicit
train_loss	7.10696	1950	82	explicit
train_loss	7.39906	1975	83	explicit
train_loss	7.09184	2000	84	explicit
train_loss	7.34296	2025	85	explicit
train_loss	7.20864	2050	86	explicit
train_loss	7.03124	2075	87	explicit
train_loss	7.24229	2100	88	explicit
train_loss	7.18057	2125	89	explicit
train_loss	7.04738	2150	90	explicit
train_loss	7.13952	2175	91	explicit
train_loss	7.22	2200	92	explicit
train_loss	6.99015	2225	93	explicit
train_loss	7.37198	2250	94	explicit
train_loss	7.36783	2275	95	explicit
train_loss	7.24509	2300	96	explicit
train_loss	6.96308	2325	97	explicit

Metric	Value	Step	Line	Source
train_loss	7.05893	2350	98	explicit
train_loss	7.17767	2375	99	explicit
train_loss	7.27932	2400	100	explicit
train_loss	7.25565	2425	101	explicit
train_loss	7.47281	2450	102	explicit
train_loss	7.15229	2475	103	explicit
train_loss	7.26168	2500	104	explicit
train_loss	7.24374	2525	105	explicit
train_loss	7.00791	2550	106	explicit
train_loss	7.07704	2575	107	explicit
train_loss	7.12217	2600	108	explicit
train_loss	7.04961	2625	109	explicit
train_loss	7.03739	2650	110	explicit
train_loss	7.20498	2675	111	explicit
train_loss	7.23664	2700	112	explicit
train_loss	7.72049	2725	113	explicit
train_loss	7.14499	2750	114	explicit
train_loss	6.87047	2775	115	explicit
train_loss	7.05357	2800	116	explicit
train_loss	7.33457	2825	117	explicit
train_loss	7.08671	2850	118	explicit
train_loss	7.3422	2875	119	explicit
train_loss	7.53044	2900	120	explicit
train_loss	7.2246	2925	121	explicit
train_loss	7.25525	2950	122	explicit
train_loss	7.21727	2975	123	explicit
train_loss	7.28318	3000	124	explicit
train_loss	7.4012	3025	125	explicit
train_loss	7.07254	3050	126	explicit
train_loss	6.96925	3075	127	explicit
train_loss	7.21686	3100	128	explicit
train_loss	7.30539	3125	129	explicit
train_loss	7.12327	3150	130	explicit
train_loss	7.48341	3175	131	explicit
train_loss	7.26789	3200	132	explicit
train_loss	7.18811	3225	133	explicit
train_loss	7.4928	3235	134	explicit
steps	3235		135	explicit
peak_memory_reserved_gb	5.32812		136	explicit
mean_tokens_per_second	3028.81		137	explicit

Phase	Line	Time	CPU time	Memory	GPU
finetune	2	15:58:53	1.5 s	0.56 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	15:58:53	7467a68a635fb44e	727
1	phase finetune	15:58:53	7467a68a635fb44e	0
2	run_end	16:55:25	7467a68a635fb44e	0

Event	Path or command	Hash
read	kv_run.sh	
read	torch tune/__init__.py	
read	.github/scripts/pre_build_script.sh	
read	.github/workflows/build_docs.yaml	
read	.github/workflows/build_linux_wheels.yaml	
read	.github/workflows/export.yaml	
read	.github/workflows/gpu_test.yaml	
read	.github/workflows/lint.yaml	
read	.github/workflows/regression_test.yaml	
read	.github/workflows/rl_test.yaml	
read	.github/workflows/unit_test.yaml	
read	.pre-commit-config.yaml	
read	docs/source/conf.py	
read	docs/source/custom_directives.py	
read	pyproject.toml	
read	recipes/__init__.py	
read	recipes/configs/dev/11B_lora_multi_dataset.yaml	
read	recipes/configs/dev/3B_full_grpo.yaml	
read	recipes/configs/dev/3B_sft_for_grpo.yaml	
read	recipes/configs/dev/8B_full_experimental.yaml	
read	recipes/configs/dev/qwen3B_async_grpo.yaml	
read	recipes/configs/dev/qwen3B_sync_grpo.yaml	
read	recipes/configs/eleuther_evaluation.yaml	
read	recipes/configs/gemma/2B_full.yaml	
read	recipes/configs/gemma/2B_lora.yaml	
read	recipes/configs/gemma/2B_lora_single_device.yaml	
read	recipes/configs/gemma/2B_qlora_single_device.yaml	
read	recipes/configs/gemma/7B_full.yaml	
read	recipes/configs/gemma/7B_lora.yaml	
read	recipes/configs/gemma/7B_lora_single_device.yaml	
read	recipes/configs/gemma/7B_qlora_single_device.yaml	
read	recipes/configs/gemma/evaluation.yaml	
read	recipes/configs/gemma2/27B_full.yaml	
read	recipes/configs/gemma2/27B_lora.yaml	
read	recipes/configs/gemma2/27B_lora_single_device.yaml	
read	recipes/configs/gemma2/27B_qlora_single_device.yaml	
read	recipes/configs/gemma2/2B_full.yaml	
read	recipes/configs/gemma2/2B_lora.yaml	
read	recipes/configs/gemma2/2B_lora_single_device.yaml	
read	recipes/configs/gemma2/2B_qlora_single_device.yaml	
read	recipes/configs/gemma2/9B_full.yaml	

Event	Path or command	Hash
read	recipes/configs/gemma2/9B_lora.yaml	
read	recipes/configs/gemma2/9B_lora_single_device.yaml	
read	recipes/configs/gemma2/9B_qlora_single_device.yaml	
read	recipes/configs/generation.yaml	
read	recipes/configs/llama2/13B_full.yaml	
read	recipes/configs/llama2/13B_lora.yaml	
read	recipes/configs/llama2/13B_qlora_single_device.yaml	
read	recipes/configs/llama2/1B_full_ppo_low_memory_single_device.yaml	
read	recipes/configs/llama2/1B_qat_single_device.yaml	
read	recipes/configs/llama2/70B_lora.yaml	
read	recipes/configs/llama2/70B_qlora.yaml	
read	recipes/configs/llama2/7B_full.yaml	
read	recipes/configs/llama2/7B_full_low_memory.yaml	
read	recipes/configs/llama2/7B_lora.yaml	
read	recipes/configs/llama2/7B_lora_dpo.yaml	
read	recipes/configs/llama2/7B_lora_dpo_single_device.yaml	
read	recipes/configs/llama2/7B_lora_single_device.yaml	
read	recipes/configs/llama2/7B_qat_full.yaml	
read	recipes/configs/llama2/7B_qlora.yaml	
read	recipes/configs/llama2/7B_qlora_single_device.yaml	
read	recipes/configs/llama2/generation_v2.yaml	
read	recipes/configs/llama3/70B_full.yaml	
read	recipes/configs/llama3/70B_generation_distributed.yaml	
read	recipes/configs/llama3/70B_lora.yaml	
read	recipes/configs/llama3/8B_dora.yaml	
read	recipes/configs/llama3/8B_dora_single_device.yaml	
read	recipes/configs/llama3/8B_full.yaml	
read	recipes/configs/llama3/8B_full_single_device.yaml	
read	recipes/configs/llama3/8B_lora.yaml	
read	recipes/configs/llama3/8B_lora_single_device.yaml	
read	recipes/configs/llama3/8B_qat_full.yaml	
read	recipes/configs/llama3/8B_qat_lora.yaml	
read	recipes/configs/llama3/8B_qdora_single_device.yaml	
read	recipes/configs/llama3/8B_qlora_single_device.yaml	
read	recipes/configs/llama3_1/405B_qlora.yaml	
read	recipes/configs/llama3_1/70B_full.yaml	
read	recipes/configs/llama3_1/70B_generation_distributed.yaml	
read	recipes/configs/llama3_1/70B_lora.yaml	
read	recipes/configs/llama3_1/8B_full.yaml	
read	recipes/configs/llama3_1/8B_full_dpo.yaml	
read	recipes/configs/llama3_1/8B_full_single_device.yaml	
read	recipes/configs/llama3_1/8B_lora.yaml	
read	recipes/configs/llama3_1/8B_lora_dpo.yaml	
read	recipes/configs/llama3_1/8B_lora_dpo_single_device.yaml	
read	recipes/configs/llama3_1/8B_lora_single_device.yaml	
read	recipes/configs/llama3_1/8B_qat_full.yaml	
read	recipes/configs/llama3_1/8B_qat_lora.yaml	
read	recipes/configs/llama3_1/8B_qlora_single_device.yaml	
read	recipes/configs/llama3_1/evaluation.yaml	
read	recipes/configs/llama3_2/1B_full.yaml	

Event	Path or command	Hash
read	recipes/configs/llama3_2/1B_full_single_device.yaml	
read	recipes/configs/llama3_2/1B_lora.yaml	
read	recipes/configs/llama3_2/1B_lora_single_device.yaml	
read	recipes/configs/llama3_2/1B_qat_lora.yaml	
read	recipes/configs/llama3_2/1B_qlora_single_device.yaml	
read	recipes/configs/llama3_2/3B_full.yaml	
read	recipes/configs/llama3_2/3B_full_single_device.yaml	
read	recipes/configs/llama3_2/3B_lora.yaml	
read	recipes/configs/llama3_2/3B_lora_single_device.yaml	
read	recipes/configs/llama3_2/3B_qat_full.yaml	
read	recipes/configs/llama3_2/3B_qat_lora.yaml	
read	recipes/configs/llama3_2/3B_qlora_single_device.yaml	
read	recipes/configs/llama3_2/8B_to_1B_KD_lora_distributed.yaml	
read	recipes/configs/llama3_2/8B_to_1B_KD_lora_single_device.yaml	
read	recipes/configs/llama3_2/evaluation.yaml	
read	recipes/configs/llama3_2_vision/11B_evaluation.yaml	
read	recipes/configs/llama3_2_vision/11B_full.yaml	
read	recipes/configs/llama3_2_vision/11B_full_single_device.yaml	
read	recipes/configs/llama3_2_vision/11B_generation_v2.yaml	
read	recipes/configs/llama3_2_vision/11B_lora.yaml	
read	recipes/configs/llama3_2_vision/11B_lora_single_device.yaml	
read	recipes/configs/llama3_2_vision/11B_qlora.yaml	
read	recipes/configs/llama3_2_vision/11B_qlora_single_device.yaml	
read	recipes/configs/llama3_2_vision/90B_full.yaml	
read	recipes/configs/llama3_2_vision/90B_lora.yaml	
read	recipes/configs/llama3_2_vision/90B_qlora.yaml	
read	recipes/configs/llama3_3/70B_full.yaml	
read	recipes/configs/llama3_3/70B_full_multinode.yaml	
read	recipes/configs/llama3_3/70B_generation_distributed.yaml	
read	recipes/configs/llama3_3/70B_lora.yaml	
read	recipes/configs/llama3_3/70B_qlora.yaml	
read	recipes/configs/llama4/maverick_17B_128E_full.yaml	
read	recipes/configs/llama4/scout_17B_16E_full.yaml	
read	recipes/configs/llama4/scout_17B_16E_generation_distributed.yaml	
read	recipes/configs/llama4/scout_17B_16E_lora.yaml	
read	recipes/configs/mistral/7B_full.yaml	
read	recipes/configs/mistral/7B_full_low_memory.yaml	
read	recipes/configs/mistral/7B_full_ppo_low_memory.yaml	
read	recipes/configs/mistral/7B_lora.yaml	
read	recipes/configs/mistral/7B_lora_single_device.yaml	
read	recipes/configs/mistral/7B_qlora_single_device.yaml	
read	recipes/configs/mistral/evaluation.yaml	
read	recipes/configs/phi3/evaluation.yaml	
read	recipes/configs/phi3/mini_full.yaml	
read	recipes/configs/phi3/mini_full_low_memory.yaml	
read	recipes/configs/phi3/mini_lora.yaml	
read	recipes/configs/phi3/mini_lora_single_device.yaml	
read	recipes/configs/phi3/mini_qlora_single_device.yaml	
read	recipes/configs/phi4/14B_full.yaml	
read	recipes/configs/phi4/14B_full_low_memory.yaml	

Event	Path or command	Hash
read	recipes/configs/phi4/14B_lora.yaml	
read	recipes/configs/phi4/14B_lora_single_device.yaml	
read	recipes/configs/phi4/14B_qlora_single_device.yaml	
read	recipes/configs/phi4/evaluation.yaml	
read	recipes/configs/quantization.yaml	
read	recipes/configs/qwen2/0.5B_full.yaml	
read	recipes/configs/qwen2/0.5B_full_single_device.yaml	
read	recipes/configs/qwen2/0.5B_lora.yaml	
read	recipes/configs/qwen2/0.5B_lora_single_device.yaml	
read	recipes/configs/qwen2/1.5B_full.yaml	
read	recipes/configs/qwen2/1.5B_full_single_device.yaml	
read	recipes/configs/qwen2/1.5B_lora.yaml	
read	recipes/configs/qwen2/1.5B_lora_single_device.yaml	
read	recipes/configs/qwen2/1.5_to_0.5B_KD_lora_distributed.yaml	
read	recipes/configs/qwen2/1.5_to_0.5B_KD_lora_single_device.yaml	
read	recipes/configs/qwen2/7B_full.yaml	
read	recipes/configs/qwen2/7B_full_single_device.yaml	
read	recipes/configs/qwen2/7B_lora.yaml	
read	recipes/configs/qwen2/7B_lora_single_device.yaml	
read	recipes/configs/qwen2/evaluation.yaml	
read	recipes/configs/qwen2_5/0.5B_full.yaml	
read	recipes/configs/qwen2_5/0.5B_full_single_device.yaml	
read	recipes/configs/qwen2_5/0.5B_lora.yaml	
read	recipes/configs/qwen2_5/0.5B_lora_single_device.yaml	
read	recipes/configs/qwen2_5/1.5B_full.yaml	
read	recipes/configs/qwen2_5/1.5B_full_single_device.yaml	
read	recipes/configs/qwen2_5/1.5B_lora.yaml	
read	recipes/configs/qwen2_5/1.5B_lora_single_device.yaml	
read	recipes/configs/qwen2_5/1.5B_qat_single_device.yaml	
read	recipes/configs/qwen2_5/14B_lora_single_device.yaml	
read	recipes/configs/qwen2_5/14B_to_7B_KD_lora_single_device.yaml	
read	recipes/configs/qwen2_5/32B_lora.yaml	
read	recipes/configs/qwen2_5/3B_full.yaml	
read	recipes/configs/qwen2_5/3B_full_single_device.yaml	
read	recipes/configs/qwen2_5/3B_lora.yaml	
read	recipes/configs/qwen2_5/3B_lora_single_device.yaml	
read	recipes/configs/qwen2_5/3B_qat_single_device.yaml	
read	recipes/configs/qwen2_5/72B_lora.yaml	
read	recipes/configs/qwen2_5/7B_full.yaml	
read	recipes/configs/qwen2_5/7B_full_single_device.yaml	
read	recipes/configs/qwen2_5/7B_lora.yaml	
read	recipes/configs/qwen2_5/7B_lora_single_device.yaml	
read	recipes/configs/qwen2_5/evaluation.yaml	
read	recipes/configs/qwen3/0.6B_full.yaml	
read	recipes/configs/qwen3/0.6B_full_single_device.yaml	
read	recipes/configs/qwen3/0.6B_lora.yaml	
read	recipes/configs/qwen3/0.6B_lora_single_device.yaml	
read	recipes/configs/qwen3/0.6B_qat_full.yaml	
read	recipes/configs/qwen3/0.6B_qat_lora.yaml	
read	recipes/configs/qwen3/1.7B_full.yaml	

Event	Path or command	Hash
read	recipes/configs/qwen3/1.7B_full_single_device.yaml	
read	recipes/configs/qwen3/1.7B_lora.yaml	
read	recipes/configs/qwen3/1.7B_lora_single_device.yaml	
read	recipes/configs/qwen3/1.7B_qat_full.yaml	
read	recipes/configs/qwen3/1.7B_qat_lora.yaml	
read	recipes/configs/qwen3/14B_lora_single_device.yaml	
read	recipes/configs/qwen3/14B_to_8B_KD_lora_single_device.yaml	
read	recipes/configs/qwen3/32B_lora.yaml	
read	recipes/configs/qwen3/32B_qat_lora.yaml	
read	recipes/configs/qwen3/4B_full.yaml	
read	recipes/configs/qwen3/4B_full_single_device.yaml	
read	recipes/configs/qwen3/4B_lora.yaml	
read	recipes/configs/qwen3/4B_lora_single_device.yaml	
read	recipes/configs/qwen3/4B_qat_full.yaml	
read	recipes/configs/qwen3/4B_qat_lora.yaml	
read	recipes/configs/qwen3/8B_full.yaml	
read	recipes/configs/qwen3/8B_full_single_device.yaml	
read	recipes/configs/qwen3/8B_lora.yaml	
read	recipes/configs/qwen3/8B_lora_single_device.yaml	
read	recipes/configs/qwen3/8B_qat_full.yaml	
read	recipes/configs/qwen3/8B_qat_lora.yaml	
read	recipes/configs/qwen3/evaluation.yaml	
read	recipes/dev/7B_full_early_exit.yaml	
read	recipes/dev/async_grpo_full_finetune_distributed.py	
read	recipes/dev/early_exit_finetune_distributed.py	
read	recipes/dev/generate_v2.py	
read	recipes/dev/generate_v2_distributed.py	
read	recipes/dev/grpo_full_finetune_distributed.py	
read	recipes/dev/lora_finetune_distributed_multi_dataset.py	
read	recipes/eleuther_eval.py	
read	recipes/full_dpo_distributed.py	
read	recipes/full_finetune_distributed.py	
read	recipes/full_finetune_single_device.py	
read	recipes/generate.py	
read	recipes/knowledge_distillation_distributed.py	
read	recipes/knowledge_distillation_single_device.py	
read	recipes/lora_dpo_distributed.py	
read	recipes/lora_dpo_single_device.py	
read	recipes/lora_finetune_distributed.py	
read	recipes/lora_finetune_single_device.py	
read	recipes/ppo_full_finetune_single_device.py	
read	recipes/qat_distributed.py	
read	recipes/qat_lora_finetune_distributed.py	
read	recipes/qat_single_device.py	
read	recipes/quantize.py	
read	tests/__init__.py	
read	tests/assets/alpaca_tiny.json	
read	tests/assets/chat_tiny.json	
read	tests/assets/generation_config.json	
read	tests/assets/generation_config_gemma.json	

Event	Path or command	Hash
read	tests/assets/hh_rlhf_tiny.json	
read	tests/assets/instruct_tiny.json	
read	tests/assets/invalid_dummy_config.yaml	
read	tests/assets/stack_exchange_paired_tiny.json	
read	tests/assets/tiny_bpe_tokenizer.json	
read	tests/assets/tiny_bpe_vocab.json	
read	tests/assets/tokenizer.json	
read	tests/assets/tokenizer_config.json	
read	tests/assets/tokenizer_config_gemma.json	
read	tests/assets/tokenizer_gemma_cropped.json	
read	tests/assets/valid_dummy_config.yaml	
read	tests/assets/vocab.json	
read	tests/assets/vqa_tiny.json	
read	tests/cache_artifacts.sh	
read	tests/common.py	
read	tests/conftest.py	
read	tests/recipes/__init__.py	
read	tests/recipes/common.py	
read	tests/recipes/dev/test_async_grpo_full_finetune_distributed.py	
read	tests/recipes/dev/test_generate_v2.py	
read	tests/recipes/test_configs.py	
read	tests/recipes/test_eleuther_eval.py	
read	tests/recipes/test_full_dpo_distributed.py	
read	tests/recipes/test_full_finetune_distributed.py	
read	tests/recipes/test_full_finetune_single_device.py	
read	tests/recipes/test_knowledge_distillation_distributed.py	
read	tests/recipes/test_knowledge_distillation_single_device.py	
read	tests/recipes/test_lora_dpo_distributed.py	
read	tests/recipes/test_lora_dpo_single_device.py	
read	tests/recipes/test_lora_finetune_distributed.py	
read	tests/recipes/test_lora_finetune_single_device.py	
read	tests/recipes/test_ppo_full_finetune_single_device.py	
read	tests/recipes/test_qat_distributed.py	
read	tests/recipes/test_qat_lora_finetune_distributed.py	
read	tests/recipes/test_qat_single_device.py	
read	tests/recipes/utis.py	
read	tests/regression_tests/test_llama2_7b.py	
read	tests/test_import_recipes.py	
read	tests/test_utils.py	
read	tests/torch tune/__init__.py	
read	tests/torch tune/_cli/__init__.py	
read	tests/torch tune/_cli/test_cat.py	
read	tests/torch tune/_cli/test_cp.py	
read	tests/torch tune/_cli/test_download.py	
read	tests/torch tune/_cli/test_ls.py	
read	tests/torch tune/_cli/test_run.py	
read	tests/torch tune/_cli/test_tune.py	
read	tests/torch tune/_cli/test_validate.py	
read	tests/torch tune/config/test_config_utils.py	
read	tests/torch tune/config/test_instantiate.py	

Event	Path or command	Hash
read	tests/torch tune/config/test_parse.py	
read	tests/torch tune/config/test_validate.py	
read	tests/torch tune/data/test_collate.py	
read	tests/torch tune/data/test_data_utils.py	
read	tests/torch tune/data/test_messages.py	
read	tests/torch tune/data/test_prompt_templates.py	
read	tests/torch tune/datasets/__init__.py	
read	tests/torch tune/datasets/multimodal/test_llava_instruct_dataset.py	
read	tests/torch tune/datasets/multimodal/test_multimodal_chat_dataset.py	
read	tests/torch tune/datasets/multimodal/test_the_cauldron_dataset.py	
read	tests/torch tune/datasets/multimodal/test_vqa_dataset.py	
read	tests/torch tune/datasets/test_alpaca_dataset.py	
read	tests/torch tune/datasets/test_chat_dataset.py	
read	tests/torch tune/datasets/test_cnn_dailymail_dataset.py	
read	tests/torch tune/datasets/test_concat_dataset.py	
read	tests/torch tune/datasets/test_grammar_dataset.py	
read	tests/torch tune/datasets/test_hh_rlhf_helpful_dataset.py	
read	tests/torch tune/datasets/test_instruct_dataset.py	
read	tests/torch tune/datasets/test_packed_dataset.py	
read	tests/torch tune/datasets/test_preference_dataset.py	
read	tests/torch tune/datasets/test_samsum_dataset.py	
read	tests/torch tune/datasets/test_sft_dataset.py	
read	tests/torch tune/datasets/test_slimorca_dataset.py	
read	tests/torch tune/datasets/test_stack_exchange_paired_dataset.py	
read	tests/torch tune/datasets/test_text_completion_dataset.py	
read	tests/torch tune/datasets/test_wikitext_dataset.py	
read	tests/torch tune/dev/rl/rewards/__init__.py	
read	tests/torch tune/dev/rl/rewards/test_rewards.py	
read	tests/torch tune/dev/rl/workers/test_postprocessing.py	
read	tests/torch tune/generation/__init__.py	
read	tests/torch tune/generation/test_generation.py	
read	tests/torch tune/models/__init__.py	
read	tests/torch tune/models/clip/__init__.py	
read	tests/torch tune/models/clip/test_clip_image_transform.py	
read	tests/torch tune/models/clip/test_clip_text_encoder.py	
read	tests/torch tune/models/clip/test_clip_tokenizer.py	
read	tests/torch tune/models/clip/test_pos_embedding_interpolation.py	
read	tests/torch tune/models/clip/test_positional_embeddings.py	
read	tests/torch tune/models/flamingo/test_flamingo_decoder.py	
read	tests/torch tune/models/flamingo/test_flamingo_encoder.py	
read	tests/torch tune/models/flux/__init__.py	
read	tests/torch tune/models/flux/test_flux_autoencoder.py	
read	tests/torch tune/models/gemma/test_gemma_tokenizer.py	
read	tests/torch tune/models/llama2/scripts/__init__.py	
read	tests/torch tune/models/llama2/scripts/compare_attention.py	
read	tests/torch tune/models/llama2/scripts/compare_decoder.py	
read	tests/torch tune/models/llama2/scripts/compare_decoder_layer.py	
read	tests/torch tune/models/llama2/scripts/compare_dora.py	
read	tests/torch tune/models/llama2/scripts/compare_feed_forward.py	
read	tests/torch tune/models/llama2/scripts/compare_fused_attention.py	

Event	Path or command	Hash
read	tests/torch tune/models/llama2/scripts/compare_lora.py	
read	tests/torch tune/models/llama2/scripts/compare_lora_attention.py	
read	tests/torch tune/models/llama2/scripts/compare_lora_llama2.py	
read	tests/torch tune/models/llama2/test_llama2_prompt_template.py	
read	tests/torch tune/models/llama2/test_llama2_tokenizer.py	
read	tests/torch tune/models/llama2/test_lora_llama2.py	
read	tests/torch tune/models/llama3/test_llama3.py	
read	tests/torch tune/models/llama3/test_llama3_tokenizer.py	
read	tests/torch tune/models/llama3_1/test_position_embeddings.py	
read	tests/torch tune/models/llama3_2_vision/test_llama_vision_lora.py	
read	tests/torch tune/models/llama4/test_llama4.py	
read	tests/torch tune/models/llama4/test_llama4_tokenizer.py	
read	tests/torch tune/models/llama4/test_llama4_transform.py	
read	tests/torch tune/models/mistral/scripts/__init__.py	
read	tests/torch tune/models/mistral/scripts/compare_feed_forward.py	
read	tests/torch tune/models/mistral/scripts/compare_mistral.py	
read	tests/torch tune/models/mistral/scripts/compare_mistral_classifier.py	
read	tests/torch tune/models/mistral/scripts/mistral_reference.py	
read	tests/torch tune/models/mistral/scripts/mistral_test_config.py	
read	tests/torch tune/models/mistral/test_mistral.py	
read	tests/torch tune/models/mistral/test_mistral_classifier.py	
read	tests/torch tune/models/mistral/test_mistral_prompt_template.py	
read	tests/torch tune/models/mistral/test_mistral_tokenizer.py	
read	tests/torch tune/models/phi3/__init__.py	
read	tests/torch tune/models/phi3/test_lora_phi3.py	
read	tests/torch tune/models/phi3/test_phi3.py	
read	tests/torch tune/models/phi3/test_phi3_position_embeddings.py	
read	tests/torch tune/models/phi3/test_phi3_tokenizer.py	
read	tests/torch tune/models/phi4/__init__.py	
read	tests/torch tune/models/phi4/test_phi4_tokenizer.py	
read	tests/torch tune/models/qwen2/__init__.py	
read	tests/torch tune/models/qwen2/test_lora_qwen2.py	
read	tests/torch tune/models/qwen2/test_qwen2.py	
read	tests/torch tune/models/qwen2/test_qwen2_tokenizer.py	
read	tests/torch tune/models/qwen2_5/__init__.py	
read	tests/torch tune/models/qwen2_5/test_tokenizer.py	
read	tests/torch tune/models/qwen3/__init__.py	
read	tests/torch tune/models/qwen3/test_tokenizer.py	
read	tests/torch tune/models/t5/__init__.py	
read	tests/torch tune/models/t5/test_t5_encoder.py	
read	tests/torch tune/models/t5/test_t5_tokenizer.py	
read	tests/torch tune/modules/__init__.py	
read	tests/torch tune/modules/_export/test_attention.py	
read	tests/torch tune/modules/_export/test_export_position_embeddings.py	
read	tests/torch tune/modules/loss/test_ce_chunked_output_loss.py	
read	tests/torch tune/modules/loss/test_cross_entropy_loss.py	
read	tests/torch tune/modules/loss/test_kd_losses.py	
read	tests/torch tune/modules/low_precision/__init__.py	
read	tests/torch tune/modules/low_precision/test_nf4_dispatch_registration.py	
read	tests/torch tune/modules/low_precision/test_nf4_linear.py	

Event	Path or command	Hash
read	tests/torch tune/modules/model_fusion/__init__.py	
read	tests/torch tune/modules/model_fusion/test_deep_fusion.py	
read	tests/torch tune/modules/model_fusion/test_early_fusion.py	
read	tests/torch tune/modules/model_fusion/test_fusion_layers.py	
read	tests/torch tune/modules/model_fusion/test_fusion_utils.py	
read	tests/torch tune/modules/moe/test_experts.py	
read	tests/torch tune/modules/moe/test_moe.py	
read	tests/torch tune/modules/peft/__init__.py	
read	tests/torch tune/modules/peft/test_dora.py	
read	tests/torch tune/modules/peft/test_lora.py	
read	tests/torch tune/modules/peft/test_utils.py	
read	tests/torch tune/modules/test_attention.py	
read	tests/torch tune/modules/test_attention_utils.py	
read	tests/torch tune/modules/test_classifier.py	
read	tests/torch tune/modules/test_common_utils.py	
read	tests/torch tune/modules/test_cosine_with_warmup.py	
read	tests/torch tune/modules/test_early_exit_loss.py	
read	tests/torch tune/modules/test_embedding_utils.py	
read	tests/torch tune/modules/test_feed_forward.py	
read	tests/torch tune/modules/test_kv_cache.py	
read	tests/torch tune/modules/test_layer_dropout.py	
read	tests/torch tune/modules/test_layernorm.py	
read	tests/torch tune/modules/test_optim.py	
read	tests/torch tune/modules/test_position_embeddings.py	
read	tests/torch tune/modules/test_rms_norm.py	
read	tests/torch tune/modules/test_transformer_decoder.py	
read	tests/torch tune/modules/test_vision_transformer.py	
read	tests/torch tune/modules/test_vq_embeddings.py	
read	tests/torch tune/modules/tokenizers/test_gpt2.py	
read	tests/torch tune/modules/transforms/test_get_canvas_best_fit.py	
read	tests/torch tune/modules/transforms/test_get_inscribed_size.py	
read	tests/torch tune/modules/transforms/test_pad_dim_to_size.py	
read	tests/torch tune/modules/transforms/test_resize_with_pad.py	
read	tests/torch tune/modules/transforms/test_tile_crop.py	
read	tests/torch tune/modules/transforms/test_transforms.py	
read	tests/torch tune/modules/transforms/tokenizers/test_hf_tokenizer.py	
read	tests/torch tune/modules/transforms/tokenizers/test_sentencepiece.py	
read	tests/torch tune/modules/transforms/tokenizers/test_tiktoken.py	
read	tests/torch tune/modules/transforms/tokenizers/test_utils.py	
read	tests/torch tune/rl/test_parameter_server.py	
read	tests/torch tune/rlhf/__init__.py	
read	tests/torch tune/rlhf/loss/__init__.py	
read	tests/torch tune/rlhf/loss/test_dpo_loss.py	
read	tests/torch tune/rlhf/loss/test_ppo_loss.py	
read	tests/torch tune/rlhf/test_rewards.py	
read	tests/torch tune/rlhf/test_sequence_processing.py	
read	tests/torch tune/training/checkpointing/test_checkpoint.py	
read	tests/torch tune/training/checkpointing/test_checkpoint_utils.py	
read	tests/torch tune/training/checkpointing/test_distributed_checkpoint.py	
read	tests/torch tune/training/test_activation_offloading.py	

Event	Path or command	Hash
read	tests/torch tune/training/test_distributed.py	
read	tests/torch tune/training/test_memory.py	
read	tests/torch tune/training/test_metric_logging.py	
read	tests/torch tune/training/test_model_util.py	
read	tests/torch tune/training/test_pooling.py	
read	tests/torch tune/training/test_precision.py	
read	tests/torch tune/training/test_profiler.py	
read	tests/torch tune/training/test_quantization.py	
read	tests/torch tune/training/test_seed.py	
read	tests/torch tune/utils/__init__.py	
read	tests/torch tune/utils/test_device.py	
read	tests/torch tune/utils/test_logging.py	
read	torch tune/_cli/__init__.py	
read	torch tune/_cli/cat.py	
read	torch tune/_cli/cp.py	
read	torch tune/_cli/download.py	
read	torch tune/_cli/ls.py	
read	torch tune/_cli/run.py	
read	torch tune/_cli/subcommand.py	
read	torch tune/_cli/tune.py	
read	torch tune/_cli/validate.py	
read	torch tune/_recipe_registry.py	
read	torch tune/config/__init__.py	
read	torch tune/config/_errors.py	
read	torch tune/config/_instantiate.py	
read	torch tune/config/_parse.py	
read	torch tune/config/_utils.py	
read	torch tune/config/_validate.py	
read	torch tune/data/__init__.py	
read	torch tune/data/_collate.py	
read	torch tune/data/_common.py	
read	torch tune/data/_messages.py	
read	torch tune/data/_prompt_templates.py	
read	torch tune/data/_torchdata.py	
read	torch tune/data/_utils.py	
read	torch tune/datasets/__init__.py	
read	torch tune/datasets/_alpaca.py	
read	torch tune/datasets/_chat.py	
read	torch tune/datasets/_cnn_dailymail.py	
read	torch tune/datasets/_concat.py	
read	torch tune/datasets/_grammar.py	
read	torch tune/datasets/_hh_rlhf_helpful.py	
read	torch tune/datasets/_instruct.py	
read	torch tune/datasets/_packed.py	
read	torch tune/datasets/_preference.py	
read	torch tune/datasets/_samsum.py	
read	torch tune/datasets/_sft.py	
read	torch tune/datasets/_slimorca.py	
read	torch tune/datasets/_stack_exchange_paired.py	
read	torch tune/datasets/_text_completion.py	

Event	Path or command	Hash
read	torchtune/datasets/_wikitext.py	
read	torchtune/datasets/multimodal/__init__.py	
read	torchtune/datasets/multimodal/_llava_instruct.py	
read	torchtune/datasets/multimodal/_multimodal.py	
read	torchtune/datasets/multimodal/_the_cauldron.py	
read	torchtune/datasets/multimodal/_vqa.py	
read	torchtune/dev/__init__.py	
read	torchtune/dev/grpo/__init__.py	
read	torchtune/dev/grpo/data.py	
read	torchtune/dev/grpo/generation.py	
read	torchtune/dev/grpo/gsm8k.py	
read	torchtune/dev/grpo/loss.py	
read	torchtune/dev/grpo/rewards.py	
read	torchtune/dev/grpo/types.py	
read	torchtune/dev/rl/__init__.py	
read	torchtune/dev/rl/data.py	
read	torchtune/dev/rl/datatypes/__init__.py	
read	torchtune/dev/rl/datatypes/request_output.py	
read	torchtune/dev/rl/datatypes/trajectory.py	
read	torchtune/dev/rl/datatypes/vllm_completion_output.py	
read	torchtune/dev/rl/generation.py	
read	torchtune/dev/rl/gsm8k.py	
read	torchtune/dev/rl/linear_grpo_loss.py	
read	torchtune/dev/rl/rewards.py	
read	torchtune/dev/rl/types.py	
read	torchtune/dev/rl/utils/__init__.py	
read	torchtune/dev/rl/utils/dist.py	
read	torchtune/dev/rl/workers/__init__.py	
read	torchtune/dev/rl/workers/datacollectors/__init__.py	
read	torchtune/dev/rl/workers/datacollectors/sync.py	
read	torchtune/dev/rl/workers/metric_logger.py	
read	torchtune/dev/rl/workers/parameter_servers/__init__.py	
read	torchtune/dev/rl/workers/parameter_servers/vllm.py	
read	torchtune/dev/rl/workers/postprocessing.py	
read	torchtune/dev/rl/workers/trainers/__init__.py	
read	torchtune/dev/rl/workers/trainers/training.py	
read	torchtune/dev/rl/workers/weight_updaters/__init__.py	
read	torchtune/dev/rl/workers/weight_updaters/weight_updater.py	
read	torchtune/generation/__init__.py	
read	torchtune/generation/_generation.py	
read	torchtune/models/__init__.py	
read	torchtune/models/clip/__init__.py	
read	torchtune/models/clip/_component_builders.py	
read	torchtune/models/clip/_convert_weights.py	
read	torchtune/models/clip/_model_builders.py	
read	torchtune/models/clip/_position_embeddings.py	
read	torchtune/models/clip/_text_encoder.py	
read	torchtune/models/clip/_tokenizer.py	
read	torchtune/models/clip/_transform.py	
read	torchtune/models/clip/inference/_transform.py	

Event	Path or command	Hash
read	torchtune/models/convert_weights.py	
read	torchtune/models/flux/__init__.py	
read	torchtune/models/flux/_autoencoder.py	
read	torchtune/models/flux/_convert_weights.py	
read	torchtune/models/flux/_model_builders.py	
read	torchtune/models/gemma/__init__.py	
read	torchtune/models/gemma/_component_builders.py	
read	torchtune/models/gemma/_model_builders.py	
read	torchtune/models/gemma/_tokenizer.py	
read	torchtune/models/gemma/gemma_norm_embedding.py	
read	torchtune/models/gemma/rms_norm.py	
read	torchtune/models/gemma2/__init__.py	
read	torchtune/models/gemma2/_attention.py	
read	torchtune/models/gemma2/_component_builders.py	
read	torchtune/models/gemma2/_convert_weights.py	
read	torchtune/models/gemma2/_model_builders.py	
read	torchtune/models/llama2/__init__.py	
read	torchtune/models/llama2/_component_builders.py	
read	torchtune/models/llama2/_model_builders.py	
read	torchtune/models/llama2/_model_utils.py	
read	torchtune/models/llama2/_prompt_template.py	
read	torchtune/models/llama2/_tokenizer.py	
read	torchtune/models/llama3/__init__.py	
read	torchtune/models/llama3/_component_builders.py	
read	torchtune/models/llama3/_model_builders.py	
read	torchtune/models/llama3/_model_utils.py	
read	torchtune/models/llama3/_parallelism.py	
read	torchtune/models/llama3/_tokenizer.py	
read	torchtune/models/llama3_1/__init__.py	
read	torchtune/models/llama3_1/_component_builders.py	
read	torchtune/models/llama3_1/_model_builders.py	
read	torchtune/models/llama3_1/_position_embeddings.py	
read	torchtune/models/llama3_2/__init__.py	
read	torchtune/models/llama3_2/_component_builders.py	
read	torchtune/models/llama3_2/_model_builders.py	
read	torchtune/models/llama3_2_vision/__init__.py	
read	torchtune/models/llama3_2_vision/_component_builders.py	
read	torchtune/models/llama3_2_vision/_convert_weights.py	
read	torchtune/models/llama3_2_vision/_encoder.py	
read	torchtune/models/llama3_2_vision/_model_builders.py	
read	torchtune/models/llama3_2_vision/_transform.py	
read	torchtune/models/llama3_3/__init__.py	
read	torchtune/models/llama3_3/_model_builders.py	
read	torchtune/models/llama4/__init__.py	
read	torchtune/models/llama4/_chunked_attention.py	
read	torchtune/models/llama4/_component_builders.py	
read	torchtune/models/llama4/_convert_weights.py	
read	torchtune/models/llama4/_encoder.py	
read	torchtune/models/llama4/_model_builders.py	
read	torchtune/models/llama4/_parallelism.py	

Event	Path or command	Hash
read	torchtune/models/llama4/_position_embeddings.py	
read	torchtune/models/llama4/_tokenizer.py	
read	torchtune/models/llama4/_transform.py	
read	torchtune/models/mistral/__init__.py	
read	torchtune/models/mistral/_component_builders.py	
read	torchtune/models/mistral/_model_builders.py	
read	torchtune/models/mistral/_prompt_template.py	
read	torchtune/models/mistral/_tokenizer.py	
read	torchtune/models/phi3/__init__.py	
read	torchtune/models/phi3/_component_builders.py	
read	torchtune/models/phi3/_convert_weights.py	
read	torchtune/models/phi3/_model_builders.py	
read	torchtune/models/phi3/_position_embeddings.py	
read	torchtune/models/phi3/_tokenizer.py	
read	torchtune/models/phi4/__init__.py	
read	torchtune/models/phi4/_model_builders.py	
read	torchtune/models/phi4/_tokenizer.py	
read	torchtune/models/qwen2/__init__.py	
read	torchtune/models/qwen2/_component_builders.py	
read	torchtune/models/qwen2/_convert_weights.py	
read	torchtune/models/qwen2/_model_builders.py	
read	torchtune/models/qwen2/_positional_embeddings.py	
read	torchtune/models/qwen2/_tokenizer.py	
read	torchtune/models/qwen2_5/__init__.py	
read	torchtune/models/qwen2_5/_model_builders.py	
read	torchtune/models/qwen2_5/_tokenizer.py	
read	torchtune/models/qwen3/__init__.py	
read	torchtune/models/qwen3/_attention.py	
read	torchtune/models/qwen3/_component_builders.py	
read	torchtune/models/qwen3/_convert_weights.py	
read	torchtune/models/qwen3/_model_builders.py	
read	torchtune/models/qwen3/_tokenizer.py	
read	torchtune/models/mol/__init__.py	
read	torchtune/models/mol/_component_builders.py	
read	torchtune/models/mol/_model_builders.py	
read	torchtune/models/t5/__init__.py	
read	torchtune/models/t5/_component_builders.py	
read	torchtune/models/t5/_convert_weights.py	
read	torchtune/models/t5/_encoder.py	
read	torchtune/models/t5/_model_builders.py	
read	torchtune/models/t5/_tokenizer.py	
read	torchtune/modules/__init__.py	
read	torchtune/modules/_export/_position_embeddings.py	
read	torchtune/modules/_export/attention.py	
read	torchtune/modules/_export/install_requirements.sh	
read	torchtune/modules/_export/kv_cache.py	
read	torchtune/modules/attention.py	
read	torchtune/modules/attention_utils.py	
read	torchtune/modules/classifier.py	
read	torchtune/modules/common_utils.py	

Event	Path or command	Hash
read	torchtune/modules/early_exit_loss.py	
read	torchtune/modules/embedding_utils.py	
read	torchtune/modules/feed_forward.py	
read	torchtune/modules/kv_cache.py	
read	torchtune/modules/layer_dropout.py	
read	torchtune/modules/layer_norm.py	
read	torchtune/modules/loss/__init__.py	
read	torchtune/modules/loss/ce_chunked_output_loss.py	
read	torchtune/modules/loss/cross_entropy_loss.py	
read	torchtune/modules/loss/kd_losses.py	
read	torchtune/modules/loss/loss_types.py	
read	torchtune/modules/low_precision/__init__.py	
read	torchtune/modules/low_precision/_register_nf4_dispatch_ops.py	
read	torchtune/modules/low_precision/nf4_linear.py	
read	torchtune/modules/model_fusion/__init__.py	
read	torchtune/modules/model_fusion/_deep_fusion.py	
read	torchtune/modules/model_fusion/_early_fusion.py	
read	torchtune/modules/model_fusion/_fusion_layers.py	
read	torchtune/modules/model_fusion/_fusion_utils.py	
read	torchtune/modules/moe/__init__.py	
read	torchtune/modules/moe/_parallelism.py	
read	torchtune/modules/moe/experts.py	
read	torchtune/modules/moe/indices.py	
read	torchtune/modules/moe/moe.py	
read	torchtune/modules/moe/utils.py	
read	torchtune/modules/optim.py	
read	torchtune/modules/peft/__init__.py	
read	torchtune/modules/peft/_utils.py	
read	torchtune/modules/peft/dora.py	
read	torchtune/modules/peft/lora.py	
read	torchtune/modules/position_embeddings.py	
read	torchtune/modules/rms_norm.py	
read	torchtune/modules/tanh_gate.py	
read	torchtune/modules/tied_linear.py	
read	torchtune/modules/tokenizers/__init__.py	
read	torchtune/modules/transformer.py	
read	torchtune/modules/transforms/__init__.py	
read	torchtune/modules/transforms/_transforms.py	
read	torchtune/modules/transforms/tokenizers/__init__.py	
read	torchtune/modules/transforms/tokenizers/_gpt2.py	
read	torchtune/modules/transforms/tokenizers/_hf_tokenizer.py	
read	torchtune/modules/transforms/tokenizers/_sentencepiece.py	
read	torchtune/modules/transforms/tokenizers/_tiktoken.py	
read	torchtune/modules/transforms/tokenizers/_utils.py	
read	torchtune/modules/transforms/vision_utils/__init__.py	
read	torchtune/modules/transforms/vision_utils/get_canvas_best_fit.py	
read	torchtune/modules/transforms/vision_utils/get_inscribed_size.py	
read	torchtune/modules/transforms/vision_utils/pad_dim_to_size.py	
read	torchtune/modules/transforms/vision_utils/resize_with_pad.py	
read	torchtune/modules/transforms/vision_utils/tile_crop.py	

Event	Path or command	Hash
read	torchtune/modules/vision_transformer.py	
read	torchtune/modules/vq_embeddings.py	
read	torchtune/recipe_interfaces.py	
read	torchtune/rlhf/__init__.py	
read	torchtune/rlhf/_types.py	
read	torchtune/rlhf/loss/__init__.py	
read	torchtune/rlhf/loss/dpo.py	
read	torchtune/rlhf/loss/ppo.py	
read	torchtune/rlhf/rewards.py	
read	torchtune/rlhf/sequence_processing.py	
read	torchtune/rlhf/utils/__init__.py	
read	torchtune/rlhf/utils/_convert_weights.py	
read	torchtune/training/__init__.py	
read	torchtune/training/_activation_offloading.py	
read	torchtune/training/_compile.py	
read	torchtune/training/_distributed.py	
read	torchtune/training/_grad_scaler.py	
read	torchtune/training/_model_util.py	
read	torchtune/training/_profiler.py	
read	torchtune/training/activations.py	
read	torchtune/training/checkpointing/__init__.py	
read	torchtune/training/checkpointing/_checkpoint_client.py	
read	torchtune/training/checkpointing/_checkpointer.py	
read	torchtune/training/checkpointing/_utils.py	
read	torchtune/training/lr_schedulers.py	
read	torchtune/training/memory.py	
read	torchtune/training/metric_logging.py	
read	torchtune/training/pooling.py	
read	torchtune/training/precision.py	
read	torchtune/training/quantization.py	
read	torchtune/training/seed.py	
read	torchtune/utils/__init__.py	
read	torchtune/utils/_device.py	
read	torchtune/utils/_import_guard.py	
read	torchtune/utils/_logging.py	
read	torchtune/utils/_version.py	
read	.gitignore	
process	/var/tmp/kv-repro/014-meta-pytorch-torchtune/env/bin/python /var/tmp/kv-re...	
process	python - /var/tmp/kv-repro/014-meta-pytorch-torchtune/data/run	

C Run history

Invocation 1, OK, 56 minutes 32 seconds, 139 output lines, started 2026-10-07T15:58:52

```
metric_hash 1c9114dbade018e9b6fefb2233521d5a881eb0841ebd69d96c06c04e508170cc
run_digest  a69e2f380ac6dc72a82af440a6e1b9bc6dcdc6cdaee9ee325e6176bee1bc7484
anchored    at run end
```

D Raw seal fields and hashes

Seal

```
data_hash      83e379ad2c69e034a106600a4146911d1837488067750279186839bc636e7a3b
signed_message_hash 769beaa30057738d82d8dab610cabac0b91b192592d49b4e7d3df3602016a7e
nonce         6f21eaa1ebedbb3cb0c7a19cba9f4a5b
signed_at     2026-10-07T16:55:27.102054+00:00
sealed_at     2026-10-07T16:55:27.161816Z
checkout_bundle cf2c2c93a72bd28c8f756591cfc27cae1df2addacd52a2e4d13c8669c4680ec4
```

Signature (base64)

```
ogfU30oyugiX1648w78PhoHCBiEMfatxIXULr9pz66DazhFhmZpGYnJ6fRICgJa0Z1dZinUPhWGrdFyv9L0pimyr
VVDgsTKUfsTb6vHWrfpu2t+KBWAAMZFHAafRkXQmZzwUJMVQCQNm7AfbuDODgsXfG1w0eVT2H3ivsd3MxEXuntt
qq0YbZReXVekIp2y5mqaJacAgn+9hSIBTnMALbE6w6CawD+ySuBgLaCk73CTko2X2cw+sivAANpqiGaFzBEq1MWp
GWptVV2A6l+b5Bc+dP6gFR/63PKRKBGPVLI6w0iWD6lrJYekJN0kirnxTv02qcdTs+fINake006ZT3R5E1mk4P2qR
QVCYimh2t6N6ZBhDPIdf+0qtobP4mK/hkQFLljDWV9WBghXxq92JykPzXNLPUEsLJ+w7XXScQN0ZbUEwDY92aXKM
brI6XvyGgp3lcnToXCu3qnp/raEiKHxW707GwmH8rMtpvRX5gXwltDxEVCKuMzMM10aVarwo+FgKnRHijBInKFmF
sxgi5zCQEpki6PyMYQ9vBLSiMe4c4BuivoEcblLCiapJWRr8JsSSxdHLokWaC8WnPPes+h0KwMdyNYxG16xthd+1
zWTDhyypQfv1K5T0IE9rCaV8F8tiljV+MD18y/2NcaKniqMae2YF93VWj0HiNXjYCiI=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlP1cjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
7qU95EFEEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNfcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPGpZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHnCy9bLgPU5annzP9gN2l
6HEMLdkXNdKQRb1rDWyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKdxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 836644cb4a68c4bf61a7e5696339a14bb36e4698bc38c41d8850147632e6a80c
stderr 8100ecc756b82fbef4b0f463795b89e7327cbaf8b9912304ee91e1e9826cbe0c
metrics 1c9114dbade018e9b6fefb2233521d5a881eb0841ebd69d96c06c04e508170cc
digest a69e2f380ac6dc72a82af440a6e1b9bc6dcdc6cdadee9ee325e6176bee1bc7484
kv_run.sh (unchanged)
pre ffb58cc7277dbd4dd2ceb19819667fdbcef2f5bc14846298db09769cf8926752
post ffb58cc7277dbd4dd2ceb19819667fdbcef2f5bc14846298db09769cf8926752
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```